

Mathias Rodríguez Castro  
Facultad de Ingeniería, Universidad de la República  
Montevideo, Uruguay  
mathiasr@fing.edu.uy

To the Editors of *Mathematical Programming Computation*

Dear Editors,

Please consider the manuscript “Pre-Export Row Scaling for Generated Mixed-Integer Programs: Mechanism Attribution and Residual-Budget Contracts” for publication in *Mathematical Programming Computation*.

Multiplying a row of a mixed-integer program by a positive number preserves it exactly while changing the numbers a solver sees. The paper asks what that change buys and what it still guarantees. The first half attributes an observed gain: holding one LP basis fixed per model across representations, geometric-mean row scaling lowers the basis condition number on every one of the 306 generated dispatch models, with median reductions of six to eight orders of magnitude across the three classes. That measures the exported file, and the paper then measures how much of it a solver’s own scaling leaves: most is recovered by the solver, a factor of a few is not, and on one class the ordering of the kernels reverses, so a condition number of an exported matrix should not be used to rank scaling policies. An advantage previously read as evidence for block metadata proves to be a coupling-kernel effect that a control using no ownership information reproduces. The second half states what a representation can guarantee: residual budgets in application units survive consistent re-emission, and the factors compatible with a budget and coefficient limits are characterized exactly, together with an incompatibility certificate, the integer-rounding term that scaling cannot remove, and exact binary64 export.

The submission fits this journal because the computations, not only their summaries, are part of what is offered: a fixed-basis audit of 306 models with an independent reconstruction of all 1,530 bases and certified rational bounds on a stress sample; 1,080 matched matrix exports; 240 operational solver runs with two matched sensitivity reruns of the same size, changing one factor at a time; and 4,752 optimizer calls across HiGHS, Gurobi and CPLEX in which every reported optimum is verified against an independent enumeration of the stored models.

The paper claims no universal scaling advantage, and several results run against the convenient reading. The completed-run gain measured on one dispatch class survives a change of seed and disappears under a change of solver. The single instance that reverses the failure-sensitive ordering behaves differently in every configuration tested. Plain coefficient centering loses verified optima relative to the unscaled model on all three solvers, attaching a residual budget to the same rule loses none relative to that rule, and whether scaling helps at all differs across the three pipelines.

Online Resource 1 holds the proof for the triangular model, the audit protocols and complete ranges, the reconstruction and experiment protocols, and the reproduction instructions. Online Resource 2 is the source and data archive: exporter, analysis and verification scripts, stored experiment data with manifests and checksums, and tests. The figures, the fixed-basis audit, the verification of the saved binary solutions and the HiGHS stress arm need no commercial licence; only the Gurobi and CPLEX arms do.

I am the sole author, received no external funding and declare no competing interests. The manuscript discloses the use of AI assistance and states that the reported numbers come from the stored runs and that I am responsible for the scientific content. It is not under consideration elsewhere and has not been published before.

Sincerely,

Mathias Rodríguez Castro